

Voice_Tut Text to speech model

Model link: -

<https://huggingface.co/mohammedaly22/VoiceTut-TTS>

Packages and dependencies: -

1. Python **3.11**
2. pip install torch torchaudio
3. pip install git+https://github.com/k2-fsa/OmniVoice.git
4. pip install voicetut-tts
5. pip install numpy scipy soundfile librosa transformers accelerate huggingface-hub safetensors pydub webdataset tensorboardX

Why do we need these packages: -

1. **PyTorch (torch) and TorchAudio (torchaudio)** – Used to run the VoiceTut-TTS model and process audio, with PyTorch providing GPU acceleration.
2. **OmniVoice** – Provides the underlying architecture and model components that VoiceTut-TTS is built on.
3. **VoiceTut-TTS (voicetut-tts)** – The main package used to load the VoiceTut model and generate Egyptian Arabic speech from text.
4. **Additional dependencies** – Packages such as NumPy, SciPy, SoundFile, Librosa, Transformers, Accelerate, Hugging Face Hub, Safetensors, Pydub, WebDataset, and TensorBoardX provide the required **numerical processing, audio processing, model loading, GPU support, model-weight downloading, and supporting functionality** needed by VoiceTut-TTS and OmniVoice.

How to run: -

```
python test_voicetut.py
```